Lab Activity 3.3: Construct form validation using Bootstrap and JavaScript

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STUDENT INFORMATION

Name	Registration No	Date
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Learning Outcomes:

a) Construct client-side form validation using JavaScript.
b) Construct Bootstrap form components for improved user experience.
c) Perform form submission and validation errors gracefully.

Step 1: Setting Up the Document Structure & Styling

Create a new file named `passport_form.html`. We will start by defining the **Purple Theme** using CSS variables and styling the passport photo container to meet the **35mm x 45mm** requirement.

HTML

```
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-
scale=1.0">
    <title>Lab 3.3: Passport Application Form</title>
    <link
href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/css/bootstrap
.min.css" rel="stylesheet">
<style>
    :root { --primary-purple: #6f42c1; --hover-purple: #59359a; --
light-purple: #f3e5f5; }
    body { background-color: var(--light-purple); padding: 40px 20px;
}
    .form-container { max-width: 800px; margin: auto; background:
white; padding: 40px; border-radius: 15px; border-top: 8px solid
var(--primary-purple); }
    .text-purple { color: var(--primary-purple); }
    .bg-purple { background-color: var(--primary-purple) !important;
}
    .btn-purple { background-color: var(--primary-purple); color:
white; }
```

```
    .btn-purple:hover { background-color: var(--hover-purple); color:
white; }
    .passport-photo { width: 3.5cm; height: 4.5cm; object-fit: cover;
border: 3px solid #f8f9fa; outline: 1px solid #dee2e6; display:
block; margin: 0 auto 10px auto; }
    .form-control:focus, .form-select:focus { border-color: var(--
primary-purple); box-shadow: 0 0 0 0.25rem rgba(111, 66, 193, 0.25);
}
    #rangeValue { background-color: var(--primary-purple); color:
white; padding: 2px 10px; border-radius: 10px; font-size: 0.9rem; }
    #successMessage { display: none; }
</style>
</head>
```

Step 2: Constructing the UI (Form Components)

Add the HTML body. This includes the **Success Alert**, **Passport Image**, **Floating Labels**, and **Input Groups**. Note the use of `novalidate` in the form tag to allow custom JavaScript handling.

HTML

```
<head>
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    <meta name="viewport" content="width=device-width, initial-
scale=1.0">
    <title>Lab 3.3: Passport Application Form</title>
    <link
href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/css/bootstrap
.min.css" rel="stylesheet">
<style>
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light-purple: #f3e5f5; }
    body { background-color: var(--light-purple); padding: 40px 20px;
}
    .form-container { max-width: 800px; margin: auto; background:
white; padding: 40px; border-radius: 15px; border-top: 8px solid
var(--primary-purple); }
    .text-purple { color: var(--primary-purple); }
    .bg-purple { background-color: var(--primary-purple) !important;
}
    .btn-purple { background-color: var(--primary-purple); color:
white; }
    .btn-purple:hover { background-color: var(--hover-purple); color:
white; }
    .passport-photo { width: 3.5cm; height: 4.5cm; object-fit: cover;
border: 3px solid #f8f9fa; outline: 1px solid #dee2e6; display:
block; margin: 0 auto 10px auto; }
    .form-control:focus, .form-select:focus { border-color: var(--
primary-purple); box-shadow: 0 0 0 0.25rem rgba(111, 66, 193, 0.25);
}
    #rangeValue { background-color: var(--primary-purple); color:
white; padding: 2px 10px; border-radius: 10px; font-size: 0.9rem; }
    #successMessage { display: none; }
</style>
```

```
</head>
```

Step 3: JavaScript Logic (Validation & Interaction)

Implement the script to handle **Range Slider** updates, **Password Matching**, and **Graceful Error Handling**.

HTML

```
<script>
    (function () {
        'use strict'

        const form = document.getElementById('passportForm');
        const successDiv = document.getElementById('successMessage');
        const resetBtn = document.getElementById('resetBtn');
        const rangeInput = document.getElementById('yearsResidency');
        const rangeValue = document.getElementById('rangeValue');

        // Update range value display
        rangeInput.addEventListener('input', function() {
            rangeValue.textContent = this.value;
        });

        // Form Validation Logic
        form.addEventListener('submit', function (event) {
            if (!form.checkValidity()) {
                event.preventDefault();
                event.stopPropagation();
                successDiv.style.display = 'none';
            } else {
                event.preventDefault();
                successDiv.style.display = 'block';
                window.scrollTo({ top: 0, behavior: 'smooth' });
            }
            form.classList.add('was-validated');
        }, false);

        // Reset Logic
        resetBtn.addEventListener('click', function() {
            form.classList.remove('was-validated');
            successDiv.style.display = 'none';
            rangeValue.textContent = "25"; // Reset range display to
default
        });

    })();
    (function () {
    'use strict'

    const form = document.getElementById('passportForm');
    const password = document.getElementById('password');
    const confirmPassword =
document.getElementById('confirmPassword');

    // Fungsi untuk menyemak padanan kata laluan
```

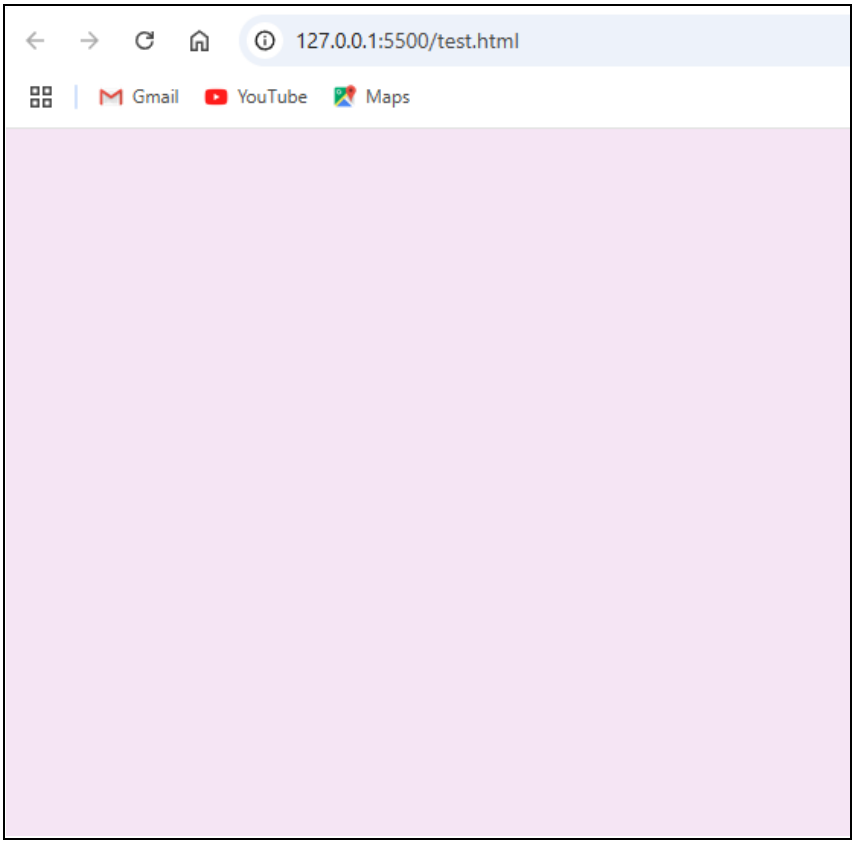
```
    function validatePassword() {
        if (password.value !== confirmPassword.value) {
            // Jika tidak sama, set ralat custom
            confirmPassword.setCustomValidity("Passwords do not
match");
        } else {
            // Jika sama, kosongkan ralat (sah)
            confirmPassword.setCustomValidity("");
        }
    }

    // Jalankan fungsi setiap kali pengguna menaip
    password.addEventListener('change', validatePassword);
    confirmPassword.addEventListener('keyup', validatePassword);

    // Kekalkan logik submit anda yang sedia ada
    form.addEventListener('submit', function (event) {
        // validatePassword dipanggil sekali lagi sebelum submit
untuk kepastian
        validatePassword();

        if (!form.checkValidity()) {
            event.preventDefault();
            event.stopPropagation();
        } else {
            event.preventDefault();
            document.getElementById('successMessage').style.display =
'block';
            window.scrollTo({ top: 0, behavior: 'smooth' });
        }
        form.classList.add('was-validated');
    }, false);

    // ... (Logik Range dan Reset anda yang lain)
})();
</script>

<script
src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/js/bootstrap.b
undle.min.js"></script>
```

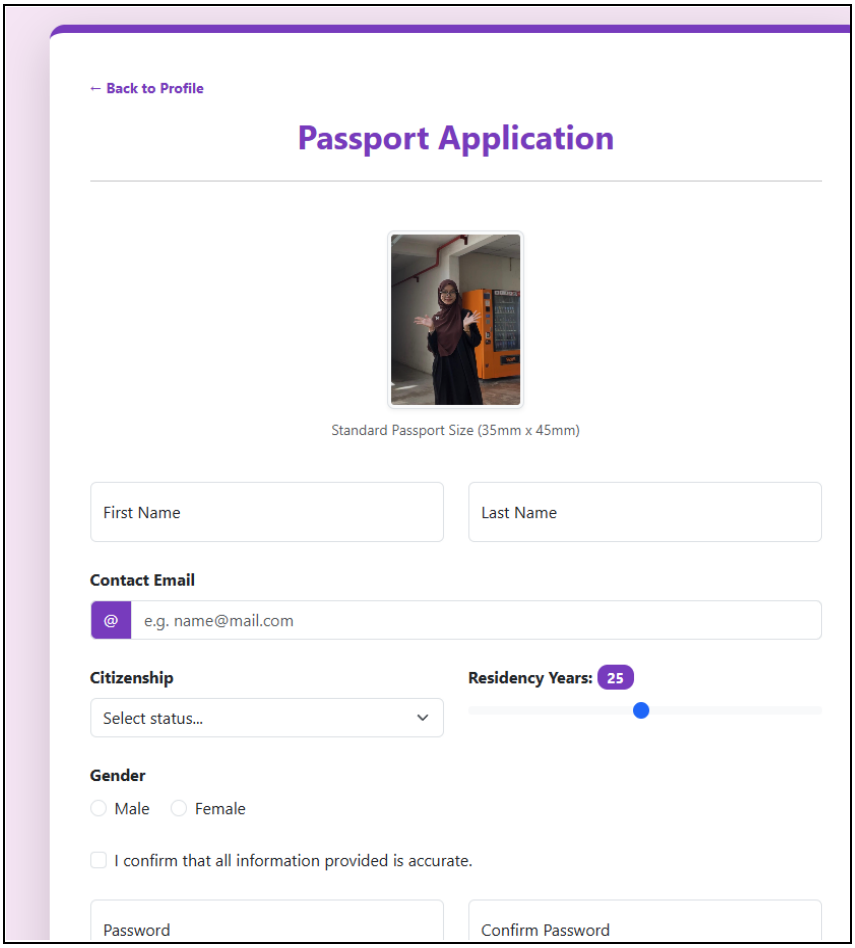
Task Recap for Students:

1. **Step 1:** Copy CSS to define the look and the specific **3.5cm x 4.5cm** image size.
2. **Step 2:** Construct the form using Bootstrap's grid and specialized components.
3. **Step 3:** Apply the JavaScript to enable live slider updates and ensure the passwords match before submission.
4. **Observation:** Notice how the `was-validated` class triggers red/green borders and messages gracefully without refreshing the page.

STEP-BY-STEP VALIDATION:

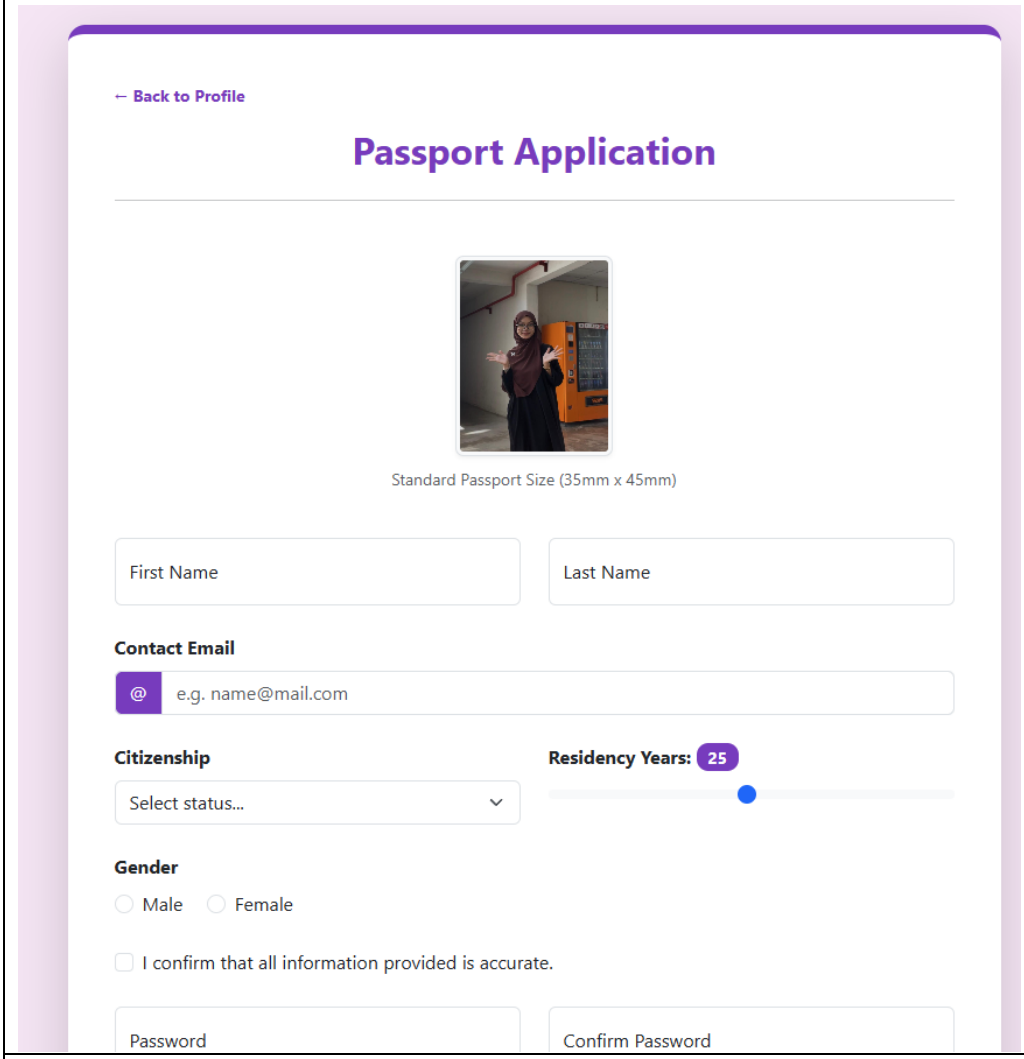
Please provide a **screenshot** for every completed step. For each image, write a short **observation** explaining the logic behind the code you just implemented. This helps solidify your mastery of Bootstrap!

Progress Step	**Screenshot of Output**	**Your Observation & Understanding**
Step 1		Nothing happen because we just only put coding of style.

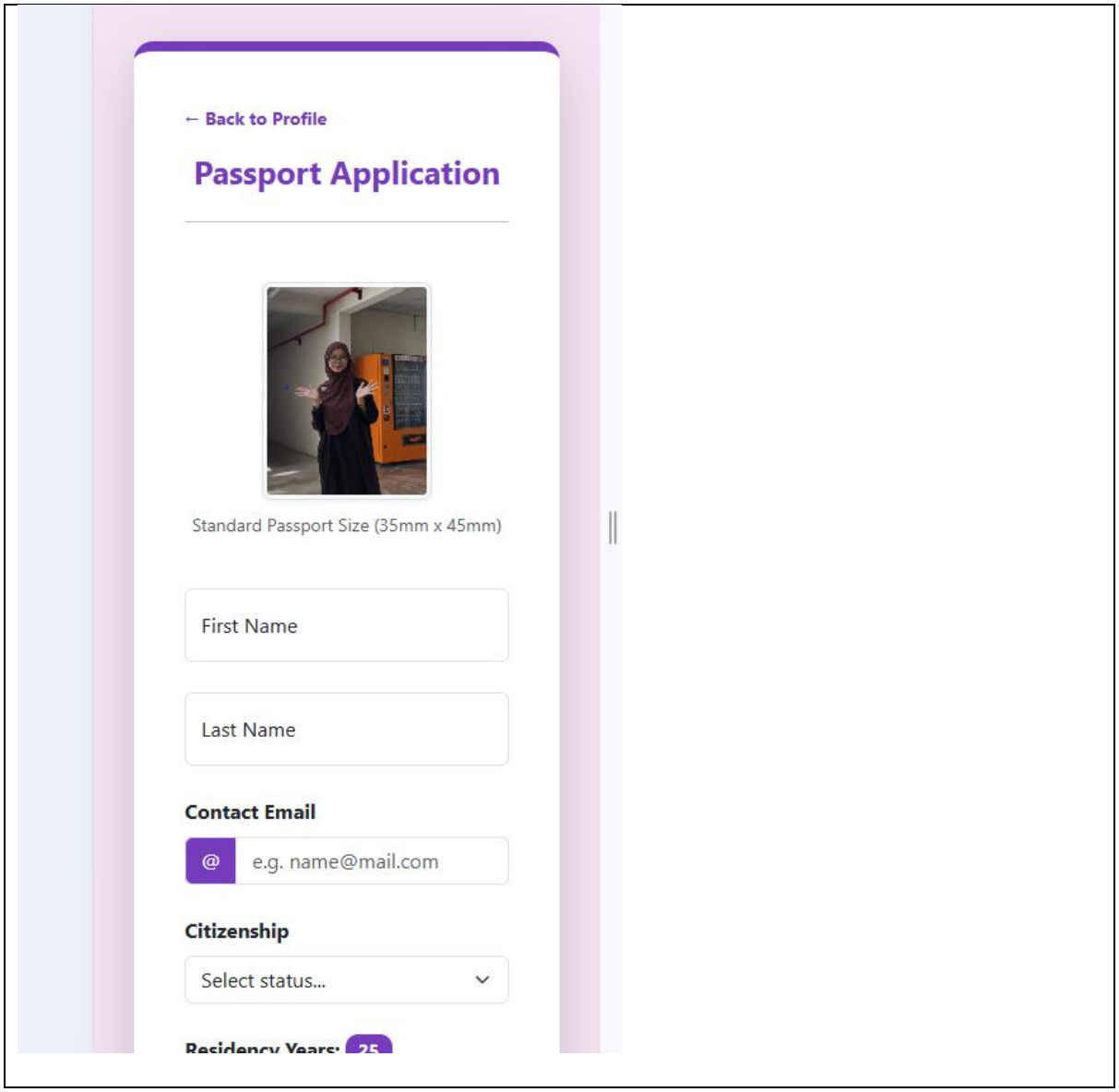
Progress Step	Screenshot of Output	Your Observation & Understanding
Step 1		Now there have an component in boostraps which is modal ,tooltips,popover, carousel.
Cont …	-	

5. SCREENSHOT OUTPUT (COMPLETE):

Desktop View



Mobile View



6. LAB SUBMISSION TASK

Ensure you have completed all steps successfully!

1. **Compress** your folder: `RegNo_Date.zip`.
2. **Report:** Create a PDF containing all screenshots and explanations: `Report_RegNo_Date.pdf`.

Happy Coding! Success is in your hands!